

ANDREW SOLOMON

Principal Salesforce Engineer — federal delivery, secure Apex and LWC architecture, Experience Cloud, and 2GP package engineering

Summary

Principal Salesforce Engineer with 15+ years of software development experience and 20+ full end-to-end Salesforce implementations across federal agencies and Fortune 500 enterprises. Founder of Arcnology, an independent Salesforce engineering practice delivering 2GP package architecture, Bulk API 2.0 data pipelines, and platform engineering standards under contract. Delivered case management, Experience Cloud, and security hardening for the Department of Veterans Affairs across the Personnel Security & Investigations Program and VBA Quality Management. Deep Apex specialist across trigger frameworks, asynchronous processing, fflib patterns, governor-limit optimization, and enterprise test architecture. Security-first approach with expertise in sharing models, persona-based testing, CRUD/FLS enforcement, and least-privilege design. Holds an active national security clearance.

Highlights

- Federal delivery for the Department of Veterans Affairs: a Salesforce case management system and Experience Cloud portal serving 10,000+ applicants and 500+ investigators for PSIP, plus least-privilege security hardening and fflib modernization for VBA QMS.
- Founder and principal engineer at Arcnology, delivering enterprise Salesforce engineering under contract across 33+ repositories of 2GP packages, GitHub Actions pipelines, and Bulk API 2.0 to Amazon S3 data pipelines.
- Security-first Apex: `Security.stripInaccessible()`, CRUD/FLS checks, sharing enforcement, `FeatureManagement.checkPermission()`, and persona-based test coverage across compliance-sensitive programs.
- Technical leadership: established Apex, LWC, Jest, and Playwright testing standards, authored engineering and DevOps documentation, and mentored engineering teams across platform, web, and AI-assisted tooling.

Technical Skills

- **Salesforce:** Apex, LWC, SOQL, SOSL, Trigger Frameworks, Async Apex, fflib, Experience Cloud, OmniStudio, 2GP Packages, Flow
- **Design:** Figma, SLDS 2, Design Tokens, IBM Carbon Design System
- **AI:** OpenAI, Anthropic SDK, AI Agents, Harness Engineering, MCP
- **Agent:** AWS Bedrock, AWS Bedrock AgentCore
- **DevOps:** GitHub Actions, SonarQube, SFDX, Vercel, Jest, Playwright
- **Frontend:** TypeScript, React, Next.js, Tailwind, GraphQL, Three.js, LWC
- **Backend:** Node, Java, Java Spring Boot, Supabase, Cloudflare Workers, Vercel Functions, REST APIs, AWS S3, AWS DynamoDB

Experience

Founder / Principal Salesforce Engineer — Arcnology

Feb 2023 — Present

- Founded an independent Salesforce engineering practice delivering architecture, 2GP package engineering, integration design, and platform standards to enterprise clients under contract.
- Own delivery end to end on each engagement: discovery, solution design, Apex and LWC implementation, DevOps pipelines, test architecture, documentation, and mentoring for client engineering teams.
- Built and scaled 2GP package pipelines across 33+ repositories, managing multiple SFDX projects in parallel across multi-tier environments.
- Architected a Salesforce Bulk API 2.0 + Amazon S3 pipeline that streams large-scale query results to S3 and generates custom object artifacts for downstream packaging and analytics.
- Developed Lightning Web Components with GraphQL and Apex service layers that integrate with Java-based connected apps for external clients.
- Established Apex and LWC testing standards, including dependency-isolated unit tests, templated Jest guidance, and Playwright browser automation for end-to-end coverage.

Senior Salesforce Developer — MagMutual

Feb 2023 — Dec 2023

- Led development of Lightning Web Components platform serving 10,000+ healthcare professionals, improving user engagement by 30%
- Architected enterprise Git workflow system with GitHub Actions, reducing deployment time by 40% across 5 development teams
- Developed Apex-backed real-time analytics dashboard processing 1M+ daily transactions with 99.9% uptime
- Implemented Apex and Jest testing suites achieving 95% code coverage and WCAG 2.1 compliance
- Built GraphQL-powered headless CMS integration, reducing content update time by 60%
- Designed component library based on IBM Carbon Design System, deployed across 3 customer-facing applications

Senior Salesforce Developer — Department of Veterans Affairs — PSIP

Apr 2022 — Jan 2023

- Designed and delivered a Salesforce case management system for VA Personnel Security & Investigations Program (PSIP)
- Built an Experience Cloud portal for 10,000+ applicants to submit forms and track case status
- Architected queue-based sharing and case assignment for 500+ investigators
- Implemented Apex-based SSO/JIT provisioning and Queueable permission set orchestration with resilient retry and error handling patterns
- Hardened Apex document access and test coverage with Security.stripInaccessible(), CRUD/FLS checks, sharing enforcement, and persona-based testing patterns

Salesforce Developer — Department of Veterans Affairs — VBA QMS

May 2021 — Mar 2022

- Refactored 15+ test classes to least-privilege, persona-based security testing

- Implemented fflib Selector, Domain, and Service patterns with CRUD/FLS enforcement across SOQL query paths
- Replaced custom Apex permission checks with FeatureManagement.checkPermission() and modernized Workflow Rules to before-save Flows
- Improved Apex test scores by 25-40 points, eliminated MIXED_DML_OPERATION failures via @testSetup, and stabilized tests around least-privilege users

Lead Frontend Engineer — CodeScience

Feb 2019 — Jan 2023

- Delivered enterprise Salesforce solutions for Fortune 500 clients including RingCentral and Ford, serving 50,000+ daily users
- Led UI and Apex architecture for 15+ projects using TypeScript, LWC, service classes, and trigger patterns, establishing core design systems
- Core contributor to Salesforce DevOps Center, improving deployment efficiency for 1000+ organizations
- Designed Apex integration and orchestration layers for AppExchange-grade managed package features while implementing Redux state management patterns that reduced component complexity by 40%
- Mentored team of 8 developers in modern JavaScript practices and Salesforce development

Senior JavaScript Developer — CNN

Aug 2018 — Feb 2019

- Developed cross-platform applications reaching 2M+ daily users across iOS, Roku, and Samsung devices
- Built real-time news delivery system using React and Node, handling 100K+ concurrent users
- Reduced application load time by 45% through performance optimization
- Led UI development for CNN's streaming platform launch

Senior Software Engineer — Cox Automotive

Jan 2014 — Feb 2018

- Architected Node microservices processing 500K+ daily automotive transactions
- Led team of 6 frontend engineers, developing component library used across 20+ applications
- Built React-based auction check-in system deployed to 100+ locations nationwide
- Developed full-stack applications using Java Spring Boot, serving 50K+ automotive dealers

Lead Frontend Developer — Look-Listen

Apr 2013 — Nov 2013

- Led frontend development for 10+ major brands including Alternative Apparel and Matchstic
- Implemented responsive design system reducing mobile bounce rate by 35%
- Established modern development workflow with Node and Sass, improving team velocity by 50%
- Built social media integration platform handling 100K+ daily API calls
- Managed AWS infrastructure supporting 1M+ monthly active users

Senior Software Developer — Paper Tiger

Aug 2010 — Dec 2012

- Developed SaaS platform serving 20K+ users with PHP and JavaScript
- Implemented Google Docs API integration processing 50K+ documents monthly
- Built single-page application improving customer support efficiency by 40%

- Designed Ruby on Rails asset pipeline reducing page load times by 60%

Education

B.Sc. Computer Science — Albany State University

Aug 2005 — May 2010

- Graduated with 3.8 GPA, focus on algorithms and information security
- Led development of security research platform used by 500+ students
- Conducted DNA sequencing research using Perl and BioPerl
- Published paper on information assurance in undergraduate research journal